

Supplementary Methods

S1. Differential Expression Analysis

Interaction model specification

Differential expression was performed using DESeq2 (v1.38.3) [1] with a full factorial interaction model that decomposes transcriptomic variation into light effects, condition (spaceflight) effects, and their interaction:

Design formula: `~ light + condition + light:condition`

Factor levels and reference categories:

Factor	Levels	Reference
Light	Red, Blue	Red
Condition	Ground, Flight	Ground

The interaction term `light:condition` captures genes where the spaceflight effect differs between red and blue light conditions. The coefficient `lightBlue.conditionFlight` represents the difference in spaceflight response between blue and red light.

Pre-filtering: Genes with fewer than 10 total counts across all samples were removed prior to analysis. After filtering, 20,157 genes remained in the leaf dataset and 21,158 in the root dataset.

Dispersion estimation: DESeq2's default parametric fit was used for gene-wise dispersion estimates, with shrinkage toward the fitted trend line to improve stability for low-count genes.

Statistical test: Wald test with Benjamini-Hochberg false discovery rate (FDR) correction for multiple testing. DESeq2's independent filtering (based on baseMean) was applied to maximize detection power.

Log-fold-change shrinkage: Adaptive shrinkage (ashr, v2.2-63) [2] was applied to all LFC estimates to reduce the impact of high-variance LFC estimates from low-count genes. The shrunk LFC values were used for all downstream analyses and visualizations.

Significance thresholds:

Effect type	Adjusted p-value	Absolute shrunk LFC	Rationale
	< 0.05	≥ 1.0	Standard genome-wide threshold

Effect type	Adjusted p-value	Absolute shrunk LFC	Rationale
Main effects (Light, Condition)			
Interaction	< 0.1	≥ 0.5	Relaxed per factorial design convention; interaction effects are inherently smaller and biologically meaningful even at modest effect sizes

The relaxed interaction threshold follows the convention for factorial designs, where interaction effects are typically smaller than main effects and detecting them requires a less stringent cutoff to avoid excessive Type II error [3]. At the strict threshold ($p_{adj} < 0.05$, $|LFC| \geq 1$), 2,894 leaf interaction genes remain significant.

Expressed gene universe: 21,786 genes passed the count filter in at least one tissue and were used as the background for all enrichment analyses.

Complete R code

```
library(DESeq2)
library(ashr)

# Load count matrix and sample metadata
# counts: gene x sample matrix of raw counts
# coldata: sample x factor data frame with columns: light, condition

# Create DESeqDataSet with interaction design
dds <- DESeqDataSetFromMatrix(countData = counts,
                              colData = coldata,
                              design = ~ light + condition + light:condition)

# Set reference levels
dds$light <- relevel(dds$light, ref = "Red")
dds$condition <- relevel(dds$condition, ref = "Ground")

# Pre-filter low-count genes
keep <- rowSums(counts(dds)) >= 10
dds <- dds[keep,]

# Run DESeq2 pipeline
dds <- DESeq(dds)

# Extract results for each effect
# Main effect: Light (Blue vs Red)
res_light <- results(dds, name = "light_Blue_vs_Red")

# Main effect: Condition (Flight vs Ground)
res_condition <- results(dds, name = "condition_Flight_vs_Ground")
```

```
# Interaction: Light × Condition
res_interaction <- results(dds, name = "lightBlue.conditionFlight")

# Apply LFC shrinkage with ashR
res_light_shrunk <- lfcShrink(dds, coef = "light_Blue_vs_Red", type = "ashr")
res_condition_shrunk <- lfcShrink(dds, coef = "condition_Flight_vs_Ground", type = "ashr")
res_interaction_shrunk <- lfcShrink(dds, coef = "lightBlue.conditionFlight", type = "ashr")

# Classify significant genes
# Main effects: padj < 0.05, |shrunk LFC| >= 1
# Interaction: padj < 0.1, |shrunk LFC| >= 0.5
sig_light <- which(res_light_shrunk$padj < 0.05 &
                  abs(res_light_shrunk$log2FoldChange) >= 1)
sig_interaction <- which(res_interaction_shrunk$padj < 0.1 &
                        abs(res_interaction_shrunk$log2FoldChange) >= 0.5)
```

Sample sizes

Tissue	Ground Red	Ground Blue	Flight Red	Flight Blue	Total
Leaf	6	6	5	4	21
Root	3	3	5	4	15
Combined	9	9	10	8	36

Note: The combined analysis pools leaf and root samples with tissue as an implicit blocking factor.

S2. Functional Enrichment Analysis

Gene Ontology and KEGG enrichment

Functional enrichment was performed using clusterProfiler (v4.10.0) [4] with the following parameters:

enrichGO (Gene Ontology):

Parameter	Value
ont	"BP" (Biological Process)
pvalueCutoff	0.1
qvalueCutoff	0.2
minGSSize	5
maxGSSize	500

Parameter	Value
pAdjustMethod	"BH" (Benjamini-Hochberg)

enrichKEGG (KEGG pathways):

Parameter	Value
organism	"sly" (Solanum lycopersicum)
keyType	"kegg"
pvalueCutoff	0.1
qvalueCutoff	0.2
minGSSize	5
maxGSSize	500
pAdjustMethod	"BH"

Background universe: All expressed genes with Entrez ID mappings (N = 16,046 of 21,786 expressed genes). Gene ID conversion from Solyc identifiers to Entrez IDs was performed using the mapping file derived from Ensembl Plants Release 109 / ITAG 4.0, yielding 16,046 unique Solyc-to-Entrez mappings. Mapping rates ranged from 59.7% to 75.0% across DEG sets.

DEG set stratification: Enrichment was computed separately for each tissue × effect × direction combination: - Leaf: light-up, light-down, condition-up, condition-down - Root: light-up, light-down, condition-up, condition-down

Interaction DEGs were further stratified by positive (stronger spaceflight effect under blue light) and negative (stronger effect under red light) directions.

S3. Cell-Type Deconvolution

Leaf cell-type mapping

Single-cell RNA-seq cluster annotations from Yue et al. [5] were used to map bulk DEGs to leaf cell types. The published UMAP clusters were annotated into six cell-type groups:

Cell type	Clusters	Description
Mesophyll	4, 5, 6, 7, 8	Photosynthetic parenchyma
Epidermis	1, 2, 10, 11	Surface cell layer
Vascular	3	Conducting tissue

Cell type	Clusters	Description
Proliferating	0	Actively dividing cells
Trichome	9	Glandular trichomes
Uncharacterized	12	Unassigned cluster

Marker genes for each cluster were obtained from the published supplementary data. Cell-type overlap was tested using Fisher's exact test (one-sided, greater) with the 21,786 expressed genes as the background universe.

Root cell-type mapping

TRAP-seq cell-type marker genes from Kajala et al. [6] were used for root deconvolution. Eleven root cell types were represented:

Cell type group	Individual cell types
Epidermis	Epidermis
Exodermis	Exodermis
Cortex	Cortex, Cortex_ACT2, Meristematic_Cortex
Endodermis	Endodermis
Vascular	Phloem, Vascular, Vascular_WOX4
Xylem	Xylem

Fisher's exact test parameters:

Parameter	Value
Alternative	"greater" (one-sided)
Background	N = 21,786 expressed genes
Multiple testing correction	Benjamini-Hochberg across all cell-type × effect × direction tests

Radial ordering: Cell types were ordered from outer to inner: Epidermis → Exodermis → Cortex → Cortex_ACT2 → Meristematic_Cortex → Endodermis → Phloem → Vascular → Vascular_WOX4 → Xylem. This ordering reflects the anatomical arrangement from the root periphery to the stele.

S4. Per-Cell-Type Pathway Enrichment

Stratification scheme

To identify cell-type-specific pathway responses, DEG sets were stratified by cell type and regulation direction before KEGG enrichment analysis. For each tissue × effect × cell-type × direction combination, the overlapping genes between the DEG set and the cell-type marker gene set were tested for KEGG pathway enrichment.

DEG set naming convention: `tissue_effect_celltype_direction` - Example:

`root_cond_Cortex_ACT2_up` = root condition-up genes that are also Cortex_ACT2 markers

KEGG enrichment parameters: Same as described in S2 (clusterProfiler `enrichKEGG` with `pvalueCutoff` = 0.1, `qvalueCutoff` = 0.2, `minGSSize` = 5, `maxGSSize` = 500).

Minimum overlap threshold: Cell-type × direction combinations with fewer than 5 overlapping genes were excluded from enrichment analysis.

Direction-stratified enrichment

For each cell type, enrichment was computed separately for upregulated and downregulated DEGs. This direction stratification enables detection of pathways that are specifically enriched in one regulatory direction within a cell type, which would be masked in a combined (up + down) analysis.

S5. PhysioSpace Stress-Pattern Mapping

Ortholog mapping

Tomato (*Solanum lycopersicum*) genes were mapped to *Arabidopsis thaliana* orthologs using Ensembl Compara (Release 109) [7] via the biomaRt R package (v2.58.2). One-to-one orthologs were used exclusively to ensure unambiguous cross-species transfer of stress signatures. Of 21,786 expressed tomato genes, 14,847 (68.2%) had one-to-one *Arabidopsis* orthologs, providing the input gene universe for PhysioSpace projection.

Reference stress spaces

Three pre-computed *Arabidopsis* stress spaces from the Plant PhysioSpace repository [6] were used as reference frameworks:

PhysioScore computation

PhysioSpace projection was performed using the R package `PhysioSpaceMethods` (v0.2.0) [5] with the `calculatePhysioMap` function. For each sample, the ranked $\log_2$ fold-change vector (treatment vs. reference) was projected onto each stress space to produce a

PhysioScore per axis, quantifying the similarity between the sample's transcriptional profile and each reference stress response.

Input preparation: Log₂ fold changes were computed per sample as the difference between each sample's normalized expression and the group mean of ground-control samples under the same light condition. Only genes with one-to-one *Arabidopsis* orthologs were used.

Projection parameters:

Space	Axes	Description
AT_Stress_Space	20	Individual abiotic and biotic stress conditions (e.g., Drought, Heat, Biotic, Light, UV)
AT_Stress_Meta	15	Meta-grouped stress categories combining related conditions (e.g., DrouSaTic = Drought + Salt + Osmotic; BioMone = Biotic + Hormone; LighUV = Light + UV)
AT_RNAseq	7	Combined stress experiments from public RNA-seq datasets (e.g., Biotic.Drought, Drought.Biotic)

Group-level contrasts

From the per-sample PhysioScore matrices, group means were computed for each of the four experimental groups (Flight_Red, Flight_Blue, Ground_Red, Ground_Blue). Three contrast types were derived:

1. **Flight effect** (per light condition): Flight group mean – Ground group mean under the same light condition. Positive values indicate stress-axis activation by spaceflight.
2. **Light effect** (per condition): Blue group mean – Red group mean under the same condition. Positive values indicate stress-axis activation by blue light.
3. **Interaction:** (Flight_Blue – Flight_Red) – (Ground_Blue – Ground_Red). Non-zero values indicate that the light modulation of the spaceflight response differs between flight and ground conditions.

Cell-type PhysioScores

Cell-type-specific PhysioScores were computed by projecting the mean log₂ fold-change profile of each cell-type marker gene set onto the three stress spaces. Four cell types were analyzed: leaf Epidermis, leaf Mesophyll, root Cortex (ACT2), and root Exodermis. Marker gene sets were derived from published single-cell atlases [8, 9] as described in Section S4.

Significance assessment

PhysioScores are bounded similarity metrics (range approximately -100 to $+100$) rather than statistical test statistics. Following the PhysioSpace convention [5, 6], axes with $|\text{PhysioScore}| \geq 10$ are considered biologically meaningful, and axes with $|\text{PhysioScore}| \geq 25$ are considered strong signals. Interaction terms with $|\text{PhysioScore}| \geq 15$ are considered notable, as interaction effects are inherently smaller than main effects.

S6. Software Versions

Parameter	Value	Rationale
Space	AT_Stress_Space, AT_Stress_Meta, AT_RNAseq	Multi-resolution stress characterization
Gene selection	One-to-one orthologs only	Unambiguous cross-species mapping
Ranking method	Default (PhysioSpaceMethods)	Rank-based projection is robust to scale differences

Software	Version	Purpose
R	4.4.2	Statistical computing environment
DESeq2	1.38.3	Differential expression analysis
ashr	2.2-63	Adaptive LFC shrinkage
clusterProfiler	4.10.0	Functional enrichment analysis
biomaRt	2.58.2	Gene ID mapping (Ensembl)
Ensembl Plants	Release 109	Reference genome annotations
ITAG	4.0	<i>S. lycopersicum</i> genome annotation
KEGG	sly (tomato)	Pathway database
Python	3.11	Visualization and data processing
matplotlib	3.8	Publication-quality figure generation
seaborn	0.13	Statistical data visualization
scipy	1.11	Statistical tests (Pearson correlation)
numpy	1.24	Numerical computing
pandas	2.0	Data manipulation
weasyprint	69.0	HTML to PDF conversion

Software	Version	Purpose
PhysioSpaceMethods	0.2.0	PhysioSpace stress-pattern projection
biomaRt	2.58.2	Ensembl ortholog mapping (also used in S1)

References

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